

XI UJIMQIN QI, China

WMO: 540120

Lat: 44.5703N

Lon: 117.6273E

Elev: 3271

StdP: 13.04

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.0	-18.7	-29.8	1.2	-22.0	-25.3	1.5	-17.3	29.2	8.8	26.6	10.9	4.3	270	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	88.6	62.7	84.8	61.9	81.7	61.0	68.0	79.5	66.2	77.9	64.6	75.9	8.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.4	102.4	73.6	62.5	95.5	70.9	60.7	89.5	70.0	34.5	79.3	32.9	77.6	31.5	76.0	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.1	20.7	17.9	-29.0	93.9	5.7	3.8	-33.1	96.6	-36.4	98.8	-39.6	100.9	-43.7	103.7		
(5)			-29.1	70.5	5.7	1.9	-33.2	71.8	-36.6	72.9	-39.8	74.0	-44.0	75.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	-1.4	6.2	21.9	40.2	53.1	63.4	69.4	65.6	54.3	38.1	18.6	3.4		
	DBStd	26.21	9.22	10.76	11.07	9.87	8.71	6.86	5.48	5.96	8.15	9.65	11.93	9.64		
	HDD50	6899	1595	1227	872	322	65	1	0	0	48	383	941	1445		
	HDD65	10829	2060	1647	1336	746	384	109	17	62	333	834	1391	1910		
	CDD50	1864	0	0	1	27	160	402	600	484	175	15	0	0		
	CDD65	318	0	0	0	1	14	60	152	81	10	0	0	0		
	CDH74	3558	0	0	0	21	256	793	1456	872	157	3	0	0		
	CDH80	1079	0	0	0	2	69	239	492	254	23	0	0	0		
(14) Wind	WSAvg	7.4	8.0	7.7	8.2	8.8	8.7	6.4	5.9	5.7	6.3	7.3	7.8	8.1		
(15) Precipitation	PrecAvg	13.1	0.1	0.1	0.2	0.4	1.1	2.0	3.9	3.1	1.3	0.6	0.2	0.1		
	PrecMax	21.5	0.4	0.3	0.6	1.9	3.1	4.2	8.2	6.7	3.0	2.2	0.7	0.3		
	PrecMin	7.4	0.0	0.0	0.0	0.0	0.2	0.4	1.0	0.8	0.2	0.0	0.0	0.0		
	PrecStd	3.2	0.1	0.1	0.2	0.4	0.6	1.0	1.6	1.6	0.8	0.6	0.2	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.5	40.0	59.4	76.9	87.1	91.1	94.4	92.5	83.3	71.4	53.9	35.5		
		MCWB	23.1	31.0	40.2	49.7	55.9	60.6	65.5	64.5	58.1	50.2	38.7	28.7		
	2%	DB	22.7	34.6	51.1	69.5	80.9	86.1	89.2	86.3	78.7	65.5	46.8	27.4		
		MCWB	18.6	27.8	37.0	46.0	53.6	59.6	65.2	63.2	56.3	47.8	35.4	22.8		
	5%	DB	18.3	29.7	45.4	63.9	75.7	82.2	85.2	82.1	74.6	60.0	41.4	22.6		
		MCWB	15.2	24.0	33.5	43.5	51.6	59.1	64.4	61.9	54.9	44.6	32.3	19.1		
	10%	DB	14.1	24.5	40.1	59.1	71.0	78.6	81.9	78.5	70.6	55.1	36.6	18.4		
		MCWB	11.5	20.1	30.6	41.4	49.9	58.1	63.7	61.0	53.6	41.9	29.7	15.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.9	32.1	41.7	51.1	58.2	64.8	71.1	69.9	63.3	52.5	40.2	28.6		
		MCDB	28.2	37.9	57.3	74.7	78.0	80.2	82.7	84.0	76.8	67.6	51.7	35.1		
	2%	WB	18.6	27.9	37.4	47.3	55.7	63.1	69.0	67.2	59.4	48.8	35.7	22.7		
		MCDB	22.3	34.7	50.3	66.2	75.1	76.7	80.6	78.7	71.4	62.5	45.7	27.2		
	5%	WB	15.2	24.1	33.9	44.6	53.5	61.6	67.3	65.3	57.2	45.8	32.7	19.1		
		MCDB	18.3	29.7	44.7	61.7	70.8	74.9	78.8	75.6	69.9	58.3	41.1	22.4		
	10%	WB	11.6	20.1	30.6	42.0	51.3	60.2	65.7	63.6	55.1	42.8	29.5	15.5		
		MCDB	14.1	24.5	39.8	58.1	68.3	73.4	77.6	73.5	67.6	54.2	36.5	18.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	21.3	21.8	23.9	24.5	22.4	20.3	21.9	24.4	22.6	19.6	17.8		
		MCDBR	24.0	26.2	29.2	32.0	32.4	29.0	26.1	27.5	29.5	30.1	24.8	22.4		
		MCWBR	20.8	20.8	18.2	15.9	13.7	10.2	8.8	10.1	12.5	16.1	16.2	19.0		
	5% WB	MCDBR	24.0	26.2	28.0	29.3	26.5	21.1	20.5	20.0	23.2	27.9	24.3	22.1		
		MCWBR	21.1	21.0	17.8	15.6	12.1	8.6	8.4	8.8	11.2	15.8	16.4	19.0		
(40) Clear-Sky Solar Irradiance	taub		0.263	0.295	0.344	0.404	0.471	0.492	0.448	0.422	0.375	0.340	0.292	0.270		
	taud		2.294	2.220	2.138	2.030	1.897	1.951	2.146	2.184	2.243	2.249	2.276	2.258		
	Ebn at Noon		273	281	280	271	255	250	260	263	267	260	256	255		
	Edh at Noon		25	33	42	51	60	57	47	43	37	32	26	24		
(44) All-Sky Solar Radiation	RadAvg		682	1053	1470	1808	1993	1935	1912	1808	1507	1084	719	530		
	RadStd		54	65	90	119	130	158	99	62	89	90	53	69		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	-3.58	-3.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	-2.86	N/A	N/A	-1.01	N/A	N/A	+122	N/A

Nomenclature: See separate page